NITTE (Deemed to be University) | NMAM INSTITUTE OF TECHNOLOGY

Finite Loop
Inspire the Rest

HACKFEST - 2026

Team Name: Team NPC (Not Paid Coders)

College Name: Siddaganga Institute of Technology, Karnataka

Track: FinTech

Problem Statement: India's 63M+ GST-registered SMBs juggle compliance, expenses, cashflow, and taxes across disconnected tools leading to missed ITC, filing penalties, and zero real-time financial clarity. No existing tool is autonomous, they all wait for the owner to do the work.

Approach:

- Ingest financial data automatically from UPI/SMS, bank PDFs, invoice photos, Telegram bills, and Excel no manual entry
- Run six specialised AI agents in parallel (GST, Expense, Profit, Cashflow, Tax, Reconciliation) using LangGraph orchestration
- Instead of dashboards, surface a **decision-first home screen** telling the owner what to do, not just what happened
- Auto-generate GSTR-1 and GSTR-3B drafts, flag mismatches before filing, and predict cash shortfalls up to 30 days ahead
- Built for how Indian SMBs actually operate Hindi voice input, Telegram bill ingestion, UPI SMS parsing

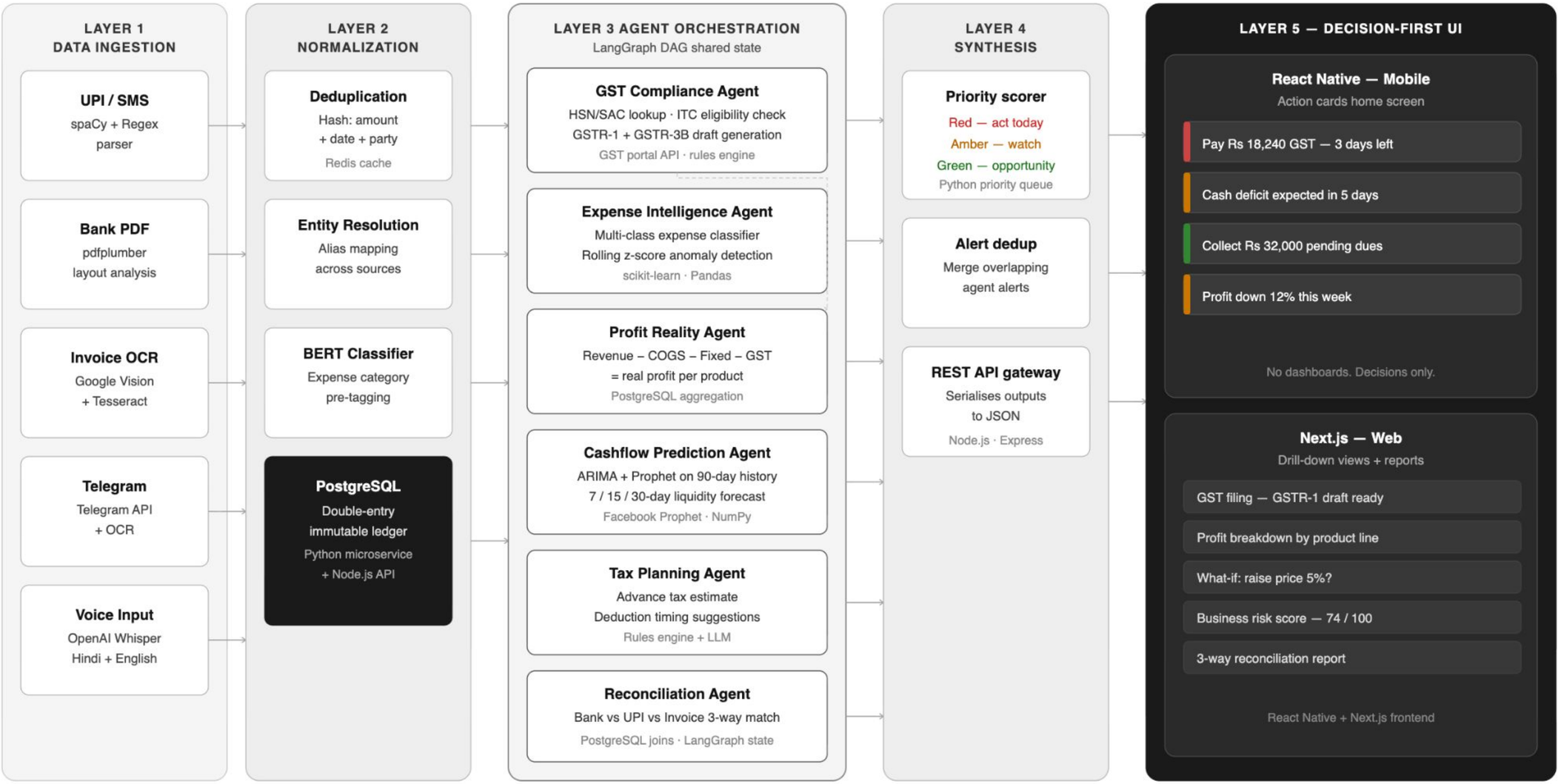
Technology Stack:

- **Frontend:** React Native (mobile) + Next.js (web)
- **Backend:** Node.js (API layer) + Python microservices
- **Agent Orchestration:** LangGraph (multi-agent, shared state)
- **OCR:** Google Vision API + Tesseract
- **Voice (NLP):** OpenAI Whisper Hindi + English
- **GST Logic:** Custom deterministic rules engine (HSN/SAC lookup, slab classification)
- **ML:** Fine-tuned expense classifier + Prophet model for cashflow forecasting
- **Database:** PostgreSQL (immutable double-entry ledger)
- **Integrations:** Telegram API, UPI SMS parser, GST portal API

Use Cases:

- **Kirana shop owner** says "Kal ₹5,000 ka maal aaya" → auto-logged, classified, reflected in cash-flow
- **Freelancer** scans a vendor invoice → GST liability calculated, GSTR-1 entry created instantly
- **Trader** gets alerted: "₹45,000 pending from 3 clients for 30+ days collect now"
- **Any SMB** runs GST Error Simulation before filing → catches wrong HSN codes, blocked ITC claims → avoids penalty
- **Small manufacturer** discovers one product is loss-making after real GST + cost deduction

Architecture:



Additional Information:

What sets us apart

Tally, Vyapar, Zoho all passive. They wait for you to enter data and show you what happened. Pocket CFO Agent ingests data automatically and tells you what to do next.

Niche features

- GST Error Simulation dry-run filing before submission; catches wrong HSN codes, blocked ITC, mismatches before penalties hit
- What-If Engine "Raise price 5%?" or "Delay this purchase 30 days?" → instant profit and cashflow impact
- Business Risk Score (0–100) composite of cashflow stability, compliance streak, expense volatility; useful for loan applications
- Voice entry "Kal ₹2000 ka payment aaya" → structured ledger entry in seconds
- Telegram bill ingestion vendor sends bill on Telegram → auto-logged, OCR-parsed, GST extracted

Market opportunity

- 63M GST-registered SMBs in India
- ₹1.5 lakh crore lost annually to compliance errors and GST penalties
- 80% of SMBs still on manual bookkeeping or basic apps

Monetisation

- SaaS: ₹499–₹1,499/month per business
- B2B2B: CA firms pay per client managed on the platform

MVP demo scope

UPI SMS ingestion → GST classification agent with ITC detection → Decision-first home screen (3 live action cards) → GST Error Simulation